

Module 02: Agents — The Self-Model, Interoception, and Ego Boundaries

Learning Objectives

1. Describe how the brain constructs a **self-model** through interoceptive inference — the perception of internal bodily states.
2. Explain the role of the insular cortex and anterior cingulate cortex in maintaining the agent's sense of self.
3. Analyze clinical conditions (depersonalization, rubber hand illusion, body integrity disorder) as disruptions of the self-model.

Introduction

What makes you *you*? Cognitive science reveals that the sense of self is not a given but a construction — an ongoing inference about the boundaries, states, and capacities of your own body. Active Inference formalizes this: the **self-model** is the component of the generative model that represents the agent's own Markov Blanket. This module examines the neural substrates and clinical disruptions of this self-model.

Key Concepts

1. Interoception as Self-Inference

Interoception — the perception of internal bodily states (heartbeat, breathing, gut sensations, temperature) — is the sensory channel through which the brain maintains its model of the body. Anil Seth (2013) argues that the "self" is fundamentally an interoceptive prediction: the brain's best guess about the causes of its internal sensory signals.

The **insular cortex** (particularly the anterior insula) is the primary cortical area for interoceptive processing. It receives input from the vagus nerve, sympathetic afferents, and chemoreceptors, and integrates these signals into a coherent representation of bodily state. In

Active Inference terms, the anterior insula maintains the self-component of the generative model.

2. Body Ownership and the Rubber Hand Illusion

The **rubber hand illusion** (Botvinick & Cohen, 1998) demonstrates that body ownership is inferential. When a participant sees a rubber hand being stroked while their own hidden hand is stroked simultaneously, the brain integrates the visual and tactile signals and "reassigns" ownership to the rubber hand.

In Active Inference terms, the generative model updates its representation of the body boundary (the Markov Blanket) based on the best explanation of the multisensory evidence. The rubber hand is incorporated into the self-model because this reduces overall prediction error across modalities.

3. Clinical Disruptions of the Self-Model

- **Depersonalization disorder:** Patients report feeling detached from their own body and mental processes. Active Inference interpretation: low precision on interoceptive prediction errors — the self-model becomes uncertain and attenuated.
- **Body integrity identity disorder (BIID):** Patients feel that a healthy limb does not belong to them and may seek amputation. Interpretation: the generative model's body representation excludes the limb, creating persistent prediction error between the model and the actual body.
- **Autism spectrum:** Some individuals show heightened interoceptive awareness combined with difficulty integrating interoceptive signals into a coherent self-model, contributing to sensory overwhelm and alexithymia (difficulty identifying emotions).

Conclusion

The self is not a homunculus but an inference — a continuously updated model of the agent's own body and boundaries. Understanding the neural and clinical dimensions of this self-model is essential for connecting Active Inference to psychopathology, embodiment, and subjective experience. Module 03 examines how the self-model interfaces with the external world through perception.